Q-HEALTHGUARD

Privacy-Preserving Federated Quantum Healthcare AI

PROPOSED SOLUTION | VNQFF-02

Q-HealthGuard proposes a framework that lets multiple hospitals collaborate on a healthcare AI model without directly exchanging raw patient records. Each hospital keeps its patient data locally and trains a local machine-learning model using a quantum machine-learning approach. Instead of sending patient records, participating hospitals send model updates for collaborative aggregation.

How the Proposed Solution Works

1. Local training at hospitals. Hospital A, Hospital B, and other participating hospitals keep medical histories, test results, and disease records on their own systems. Each hospital trains its local model using its own data.

2. Protect model updates. Before updates are shared, the design proposes privacy measures such as update clipping and Differential Privacy, plus secure aggregation preparation. These measures are intended to reduce information leakage; they do not guarantee perfect privacy by themselves.

3. Global model aggregation. A central server combines the protected updates using a federated-learning approach such as Federated Averaging (FedAvg) to create a shared global model. The quantum component is explored within the local model, for example through a quantum kernel or parameterized quantum circuit, depending on the implementation.

4. Share the global model back. The global model is distributed to participating hospitals. Hospitals can use it as a starting point for future local training and healthcare prediction tasks.

Main Components

Federated Learning: coordinates learning across hospitals while raw records stay local. **Quantum Machine Learning:** investigates quantum circuits or quantum kernels for healthcare prediction. **Privacy protection:** update clipping and calibrated Differential Privacy can reduce leakage risk. **Aggregation:** combines model updates into a shared model. A secure communication or encryption layer must be implemented and tested before claiming that updates are encrypted or fully secure.

What We Aim to Achieve

The goal is to evaluate whether collaborative training can produce a useful healthcare prediction model while limiting raw-data sharing and reducing privacy risks. The prototype can simulate three hospital clients using synthetic or public data. Any claims about accuracy, privacy, or quantum advantage should be supported by experimental results.

Proposed architecture diagram



Team: Epic Engineers • Qiskit Fall Fest 2026 • Proposed design; validate through implementation and testing.